COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212-Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/15/26	**DATE SUBMITTED:**	10/15/26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

1. Create a new repository for this activity.

```
(base) nikko@Nikko:~/activity_repo/activity_11$ git remote -v
origin  https://github.com/illuminado22/activity_repo (fetch)
origin  https://github.com/illuminado22/activity_repo (push)
(base) nikko@Nikko:~/activity_repo/activity_11$ git branch
* activity-11
  main
(base) nikko@Nikko:~/activity_repo/activity_11$ git
```

2. Install Docker and enable the docker socket.
Before installing the docker inside of the control node we need to first update the package manager.
Note: update the package manager regularly

```
(base) nikko@Nikko:~$ sudo apt update
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://security.debian.org/debian-security trixie-security InRelease
Hit:4 https://download.docker.com/linux/debian trixie InRelease
Hit:5 http://ftp.debian.org/debian trixie-backports InRelease
10 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Installing the docker and enabling the docker socket

```
(base) nikko@Nikko:~$ sudo apt install -y docker.io python3-docker python3-pip
The following package was automatically installed and is no longer required:
  pigz
Use 'sudo apt autoremove' to remove it.

Installing:
  docker.io  python3-docker  python3-pip

Installing dependencies:
  containerd     libexpat1-dev     libmodule-find-perl        libpython3-dev
```

To confirm that docker is installed successfully we can use the command “docker –version”

```
(base) nikko@Nikko:~$ docker --version
Docker version 26.1.5+dfsg1, build a72d7cd
```

To see if the dockerd which is the docker daemon is running we can use “sudo systemctl status docker”

```
(base) nikko@Nikko:~$ sudo systemctl status docker
○ docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: inactive (dead) since Thu 2026-10-08 15:18:52 PST; 6min ago
   Duration: 29min 31.738s
 Invocation: 94db476533544143a27ae16617f5736f
TriggeredBy: ○ docker.socket
       Docs: https://docs.docker.com
   Main PID: 402 (code=exited, status=0/SUCCESS)
   Mem peak: 125.5M
        CPU: 7.412s
```

```
(base) nikko@Nikko:~$ sudo systemctl enable --now docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
```

```
(base) nikko@Nikko:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-08 15:31:44 PST; 1h 7min ago
 Invocation: c9bb8c84383a45de9d38de484596c8cb
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 7447 (dockerd)
      Tasks: 18
     Memory: 79.9M (peak: 111.7M)
        CPU: 1.390s
```

3. Add to Docker group to your current user.

```
(base) nikko@Nikko:~$ sudo usermod -aG docker $USER
(base) nikko@Nikko:~$ newgrp docker
(base) nikko@Nikko:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
(base) nikko@Nikko:~$
```

Newgrp docker is for applying the chagnes into the current session then check if the docker is now a member of group. To check this we can just type a docker command since if the docker is not included the $USER we have to use sudo to use the docker.

4. Create a Dockerfile to install web and DB server.

```
FROM ubuntu:22.04
ENV DEBIAN_FRONTEND=noninteractive

RUN apt udpate && install -y \
    apache2 \
    mariadb-server \
    curl \
    && rm -rf /var/lib/apt/lists/*

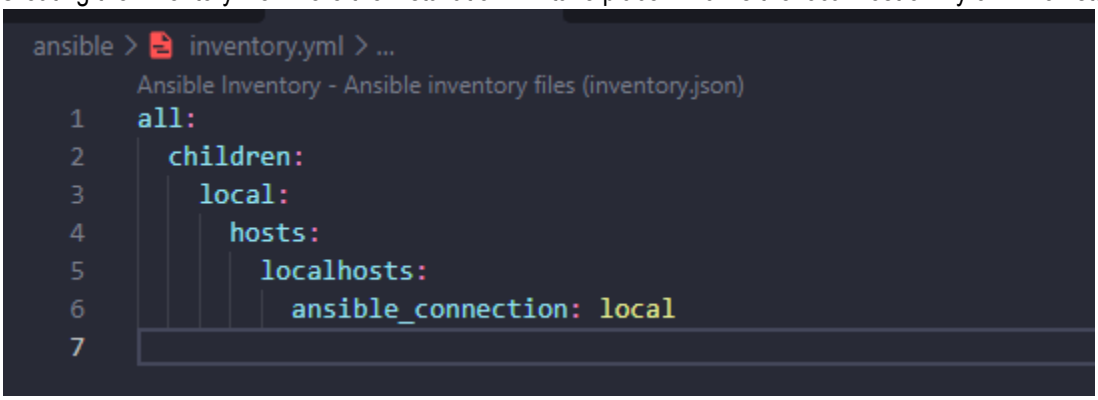
RUN echo "<h1>Web and DB server for hoa 11</h1>" > /var/www/index.html

EXPOSE 80 3306

CMD service mariadb start && apachectl -D FOREGROUND
```

5. Install and build the Dockerfile using Ansible.

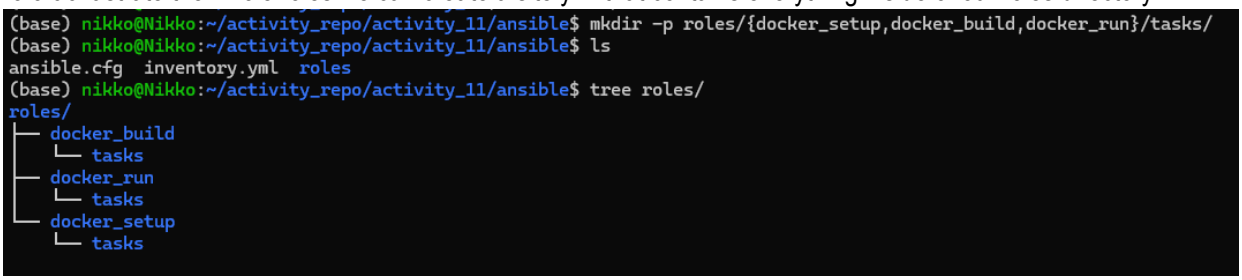
Creating the inventory file where the installation will take place:which is the local host or my own workstation.



Adding ansible.cfg so that I don't have to explicitly type the -i inventory

```
[defaults]
inventory = ./inventory.yml
```

To orachastrate the whole roles we can create a site.yml that contains everything inside of our roles directory.



We can do a recursive creation of folder in just one command using the this: “mkdir -p roles/{docker_setup,docker_build,docker_run}/tasks/”. And to see if we created the correct folder structrture we can use a tool “tree”.

After this we can now create the site.yml to orchestrate the whole role.

```
---
- name: orachastrate the whole docker setup, build, and run
  hosts: localhost
  connection: local
  become: true
  roles:
    - role: docker_setup
      tags: setup

    - role: docker_build
      tags: build

    - role: docker-run
      tags: run
```

roles/docker_setup/tasks/main.yml

```
---
- name: Install Docker and Python SDK packages
  ansible.builtin.apt:
    name:
      - docker.io
      - python3-docker
      - python3-pip
    state: present
    update_cache: yes

- name: Enable and start Docker daemon and socket
  ansible.builtin.systemd:
    name: "{{ item }}"
    state: started
    enabled: yes
  loop:
    - docker
    - docker.socket
```

```
- name: Add current user to docker group
  ansible.builtin.user:
    name: "{{ ansible_env.SUDO_USER | default(ansible_user_id) }}"
    groups: docker
    append: yes
```

roles/docker_build/tasks/main.yml

```
---
- name: Build image from Dockerfile
  community.docker.docker_image:
    name: web_db_image
    tag: latest
    build:
      path: "{{ playbook_dir }}"
    source: build
    state: present
```

roles/docker_run/tasks/main.yml

```
---
- name: Run web and DB container
  community.docker.docker_container:
    name: web_db_container
    image: web_db_image:latest
    state: started
    restart_policy: always
    published_ports:
      - "8080:80"
      - "3306:3306"
```

Confirmation of setup:

```
(base) nikko@Nikko:~/activity_repo/activity_11/ansible$ ansible-playbook -K --tags setup site.yml
ok: [localhost]

TASK [docker_setup : Install Docker and Python SDK packages] ******************************************************************************
ok: [localhost]

TASK [docker_setup : Enable and start Docker daemon and socket] ***************************************************************************
ok: [localhost] => (item=docker)
ok: [localhost] => (item=docker.socket)

TASK [docker_setup : Add current user to docker group] ************************************************************************************
ok: [localhost]

PLAY RECAP ********************************************************************************************************************************
localhost                  : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

I used tags for separate running and easy debugging. Tags is just an alias for running an specific playbook.

Run the build playbook

```
(base) nikko@Nikko:~/activity_repo/activity_11/ansible$ ansible-playbook -K --tags build site.yml
BECOME password:

PLAY [orachastrate the whole docker setup, build, and run] ********************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
ok: [localhost]

TASK [docker_build : Build image from Dockerfile] *****************************************************************************************
ok: [localhost]

PLAY RECAP ********************************************************************************************************************************
localhost                  : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Run the run playubook:

```
(base) nikko@Nikko:~/activity_repo/activity_11/ansible$ ansible-playbook -K --tags run site.yml
BECOME password:

PLAY [orachastrate the whole docker setup, build, and run] ********************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
ok: [localhost]

TASK [docker_run : Run web and DB container] **********************************************************************************************
changed: [localhost]

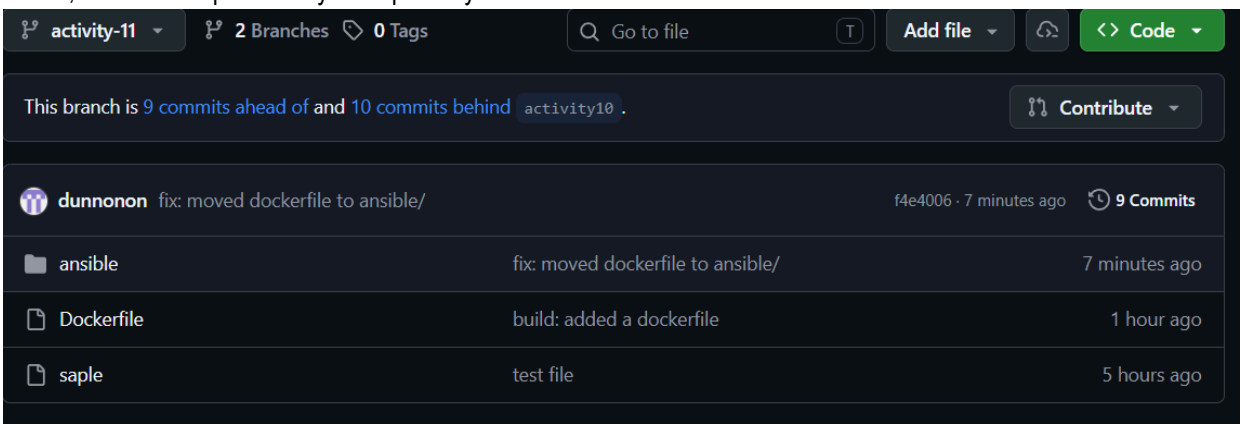
PLAY RECAP ********************************************************************************************************************************
localhost                  : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

To see if the container is running successfully we can use the command docker ps or docker ps -a

```
(base) nikko@Nikko:~/activity_repo/activity_11/ansible$ docker  ps
CONTAINER ID   IMAGE                COMMAND                  CREATED         STATUS         PORTS
                    NAMES
b10867b46b34   web_db_image:latest   "/bin/sh -c 'service…"   3 minutes ago   Up 3 minutes   0.0.0.0:3306->3306/tcp, 0.0
.0.0:8080->80/tcp   web_db_container
```

In the image above we can see that the container is running inside prot 3306.

6. Add, commit and push it to your repository.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:
Answer the following:
1. What are the benefits of implementing containerizations?
   - This benefits developers because it can transfer the dependencies into a standardized, isolated container. Because docker shares the whole OS kernel this fastens the setting up and is light weight it doesn't have to load up the whole operating system unlike VM where it takes time to boot. This can be used in CI/CD pipeline where everything needs to be fast and automated the container will be shipped into the pipeline instead of pushing everything.

Conclusion:
In conclusion, this activity demonstrated how to use docker. Installing the CLI where I can just navigate it using command instead of using the desktop. Docker is fast and light weight this is because it shares the kernel of the host OS. That's why we need to add docker to $USER because it acts like a user. We can also create multiple containers that can contain a linux as long as the host OS is linux windows for windows like that or containerized the wed/app into a single container for easy shipping.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance, explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Give my idea to AI to check if my understanding is correct and on point and code assistance for complex parts.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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